BYEONGGUK JEON

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RESEARCH INTERESTS

Video world models, neural simulation for robot policy evaluation, cross-embodiment robot learning, generative modeling (diffusion and flow matching)

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Seoul, South Korea
Ph.D. in Artificial Intelligence, Kim Jaechul Graduate School of AI (admitted) — Mar 2027 ~
M.S. in Artificial Intelligence, Kim Jaechul Graduate School of AI. Advisor: Minjoon Seo — 2025 ~ Feb 2027 (expected)

- GPA: 4.06/4.30. Research focus: video world models for robot manipulation

Korea University — Seoul, South Korea
B.S. in Computer Science and Engineering — 2018 ~ 2024

- Major GPA: 4.28/4.5, Cumulative GPA: 4.14/4.5

PUBLICATIONS (* equal contribution)

Preprints and Under Review

[P2] **RoboWorld: Fast and Reliable Neural Simulators for Generalist Robot Policy Evaluation.** **Byeongguk Jeon**, Seonghyeon Ye, JaeHyeok Doo, Sungdong Kim, Minjoon Seo, Hyungmok Son, Kimin Lee. *Under review at ICRA 2027*. [arXiv]

[P1] **SPACE: Enabling Learning from Cross-Robot Data Toward Generalist Policies.** Haeone Lee, **Byeongguk Jeon**, Suchae Jeong, Jian Kim, Kimin Lee. *arXiv 2026.* [arXiv]

Conference

[C5] **HABIT: Human-Aware Behavior and Interaction Training Dataset for Robot Manipulation.** Jaehwi Song, Suchae Jeong, **Byeongguk Jeon**, Sungdong Kim, Hyungmok Son, Kimin Lee. *NeurIPS 2026 Datasets and Benchmarks Track (Oral).* [arXiv]

[C4] **Q-Flow: Stable and Expressive Reinforcement Learning with Flow-Based Policy.** JaeHyeok Doo, **Byeongguk Jeon**, Seonghyeon Ye, Kimin Lee, Minjoon Seo. *ICML 2026.* [arXiv]

[C3] **Latent Action Pretraining from Videos.** Seonghyeon Ye*, Joel Jang*, **Byeongguk Jeon**, Sejune Joo, Jianwei Yang, Baolin Peng, Ajay Mandlekar, Reuben Tan, Yu-Wei Chao, Bill Yuchen Lin, Lars Liden, Kimin Lee, Jianfeng Gao, Luke Zettlemoyer, Dieter Fox, Minjoon Seo. *ICLR 2025.* [arXiv]

[C2] **Ask Optimal Questions: Aligning Large Language Models with Retriever's Preference in Conversational Search.** Chanwoong Yoon*, Gangwoo Kim*, **Byeongguk Jeon**, Sungdong Kim, Yohan Jo, Jaewoo Kang. *NAACL 2025 Findings.* [arXiv]

[C1] **Tree of Clarifications: Answering Ambiguous Questions with Retrieval-Augmented Large Language Models.** Gangwoo Kim, Sungdong Kim, **Byeongguk Jeon**, Joonsuk Park, Jaewoo Kang. *EMNLP 2023.* [arXiv]

Workshop

[W1] **Let's Predict Sentence by Sentence.** Hyeonbin Hwang*, **Byeongguk Jeon***, Seungone Kim, Jiyeon Kim, Hoyeon Chang, Sohee Yang, Seungpil Won, Dohaeng Lee, Youbin Ahn, Minjoon Seo. *COLM 2025 Workshop (Oral).* [arXiv]

EXPERIENCE

Config Intelligence, Researcher — Jul 2025 ~ Present

- Developed RoboWorld, an action-conditioned video world model for closed-loop evaluation of generalist robot policies (Spearman 0.933 with real-world policy rankings).
- Building a human-to-robot data conversion pipeline that uses a video world model to turn bare-hand human videos into five-finger dexterous robot data.
- Led a project on memory for robot policies.

NAVER, AI Researcher and Engineer — Jul 2024 ~ Feb 2025

LK Lab, KAIST, Research Intern (Advisor: Minjoon Seo) — Jan 2024 ~ Jun 2024

DMIS Lab, Korea University, Research Intern (Advisor: Jaewoo Kang) — Nov 2022 ~ Aug 2023

FnGuide, AI Team Intern — Jul 2022 ~ Aug 2022

SKILLS AND AWARDS

Skills: Python, PyTorch, distributed training (DDP, FSDP), Docker, Linux, Git
Awards: FnGuide Financial Big Data Contest, 4th Place; Korea University Creative Challenger Program, President's Award; Capstone Design (COSE489), 2nd Place